

DETECTION OF VOLCANIC TEMPERATURE ANOMALIES USING DRONE THERMAL SCANNING: AN APPLICATION ON HELLENIC VOLCANOES

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ABSTRACT

The Nisyros volcano located on the southeast part of the Aegean Plate boundary belongs to the Hellenic Volcanic Arc that results from the manifestation of magmatic activity by the African subducting slab beneath the Aegean overriding micro-plate. Hellenic Volcanoes are very attractive touristic locations for millions of tourists annually. Volcanic monitoring techniques may promote a sustainable touristic environment, especially with new and improved technological advantages of thermal drone scanning. A series of repeated survey flights and annual thermal mappings of an active crater, such as in Nisyros, is presented in this research. Studying changes in the craters' surface and surrounding landscape and temperature changes is essential to determine the amount of erosion, stability, and volcanic activity. Drone imaging and the use of photogrammetry mapping software are helpful tools that may reconstruct a digital elevation model of a volcano, such as the active Hellenic volcanoes including but not limited to Methana, Sousaki, Santorini, Milos, and Nisyros area on a thermal and visual spectrum. This reconstruction can then be compared and subtracted from previous years of digital elevation models and thermal footprint data, permitting the detection of land changes due to volcanic or erosion activity. The result of a series of repeated flights using optical and thermal imaging cameras at the active Hellenic Volcanoes has provided a valuable database of the thermal signature from each volcano. In this research, we present the active volcanic crater of Stephanos on Nisyros Island. Using a DJI Matrice drone with a FLIR XT thermal camera (13 mm), we could map at super high-resolution surface temperatures with less than 5 cm per pixel. Furthermore, we observe no significant changes in the surface temperature between 2019 and 2022. This methodology can also be applied to other volcanoes with annual periodic scanning to investigate normal volcanic activity.

BACKGROUND

The Hellenic Volcanic Arc (HVA) includes many active volcanic islands located in the southern Aegean Sea including but not limited to Sousaki, Santorini, Milos, and Nisyros.

The HVA arc was formed about 16 million years ago due to the subduction of the African Plate underneath the Aegean Sea Plate. As one of the most seismically active areas in the world includes recent eruptions occurred in 1888 at Nisyros and 1950 at Santorini.

Due to rapid changes in volcanic and seismic activity in this region, the volcanic islands located on the southern margin of the Aegean Sea Plate require a yearly evaluation to detect complex land changes in the area.

The evaluations of these volcanic islands are crucial to the livelihoods of civilians occupying the region since these land changes could be the result of erosion and/or volcanic activity and may pose a potential hazard to sustainable habitability.

Students and faculty members from Hofstra University in New York and the National and Kapodistrian University of Athens in Greece have applied multiple methods of volcanic investigation techniques that include drone mapping on the thermal and visual spectrum. Drone mapping has been used as a reliable method and takes place on an annual basis.

METHODS

Two different types of drones and cameras were used to take images of the Stefanos crater in Nisyros. The first drone was used to take pictures of the crater on the visual spectrum (DJI MAVIC PRO 2) while the second drone was used to take pictures on the thermal spectrum with an XT FLIR (13 mm) thermal camera.

Visual images of the surface of the Stefanos crater were taken in the middle of the day whereas thermal images of the crater were taken at night when heat from the sun is most likely to not affect the clarity of the scans.

Hundreds of these images were then processed using Global Mapper and Pix4D, a photogrammetry software for professional drone mapping to create a digital elevation (DEM) and thermal surface (TSM) models of the crater.

DEM & TSM models were then compared to the models of previous years to see if any significant changes have occurred at the crater.



(left picture): Nicholas Mead (Hofstra student) landing the visual spectrum drone after mapping the volcanic crater of Stefanos at Nisyros island during a day (nadir) scanning. (right) Jacqueline Bourgeois (Hofstra student) monitoring the thermal drone scanning the volcanic crater of Stefanos at Nisyros island during the night for mapping heat emissions.

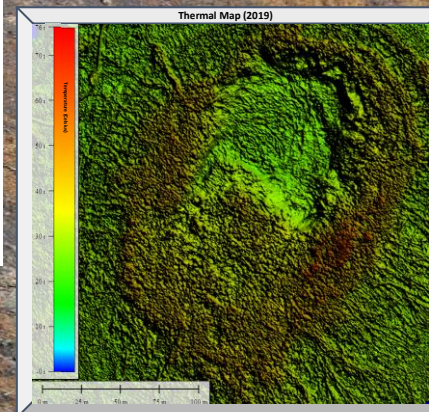


Figure 1: Thermal map made in Pix4D and reclassified in Global Mapper v.23.1 with a vertical scale showing the maximum and minimum temperatures scanned on Stefanos crater at Nisyros in 2019.

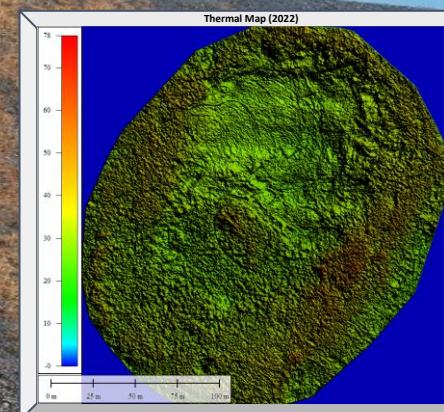


Figure 2: Thermal map made within the same parameters, equipment and methodology by scanned thermal images in 2022 (DJI Matrice M100).

RESULTS

Thermal Reconstructions after subtraction from previous years indicate that the crater seemed to be warmer in 2019 but with few exceptional higher temperatures in 2022.

There is a difference of the maximum temperature by 2 °C.

Digital Elevation Models are still in processing to compare the elevation of Stefanos from current and previous years to reveal changes at the crater and surrounding areas.

Background picture: Geoteam ascending an active volcano at the Hellenic Volcanic Arc for data and sampling collection at the Hofstra monitoring station.

CONCLUSION

Based on the results from the thermal scanning of Nisyros Stefanos crater, we can conclude that other than a slight difference in temperature, most of the heat emissions at the Stefanos crater thermal map seem to be similar.

The temperature map shows an increase of the maximum temperature by 2°C slightly but it is possible that it may be accounted an atmospheric influence that requires further investigation.

Evaluations and future assessments can be performed annually to monitor any changes of the crater in order to provide a useful analysis and keep the inhabitants of the island in a safe and sustainable environment.

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